

Properties of Physical Bodies

Below given are some of the basic properties of Physical Bodies:

Property	Description
Simulate Physics	If this property is true, then the body will use simulation, while if it is false the body won't move until it is told where to move. For the case of SkeletalMeshActors, you are required to set up physical assets that are assigned on the Skeletal Mesh Setup. While in the case of StaticMeshActors, simulating requires simple collision on Static Mesh Asset.
Mass in KG	Mass of the Body in Kilograms
Angular Damping	The Drag Force used to reduce angular movement of the given object
Linear Damping	The Drag force used to reduce the linear movement of the given object
Enable Gravity	This property decides whether an object should have the force of gravity applied to it or not.

Physics Sub-Stepping

With the use of Physics Sub-Stepping, the game developer can produce physical simulations that are more accurate as well as more stable, however, this comes at the expense of the performance of the simulation. You can enable Physics Sub-Stepping in the Project Setting -> Physics

Properties of Physics Sub-stepping

Property	Description
Sub-Stepping	This enables or disables the sub-stepping for your particular simulation project.
Substep Async	This decides whether to async physical stimulation with sub-stepping
Max Substep Delta Time	This is defined as the maximum amount of time a substep is allowed to take which is defined in seconds. So, in the case that a Full Sub-step takes 0.5 seconds to complete but the Max Sub-Step Delta Time is set at 0.25, then the given Sub-Step will be divided and executed into two equal each taking up to 0.25 seconds to execute.
Max Sub-Steps	As the name defines, it defines the maximum number of sub-steps a full physical step is allowed to be broken into.